

Experimental and Computational
Heat Transfer Group

AIR COOLING OF ELECTRONICS: A Personal Perspective 1981-2001

Alfonso Ortega

Department of Aerospace & Mechanical Engineering

The University of Arizona

Email: ortega@u.arizona.edu

Web: <http://w3.arizona.edu/~thermlab/>

2002 IEEE SEMITHERM Symposium

March 14, 2002

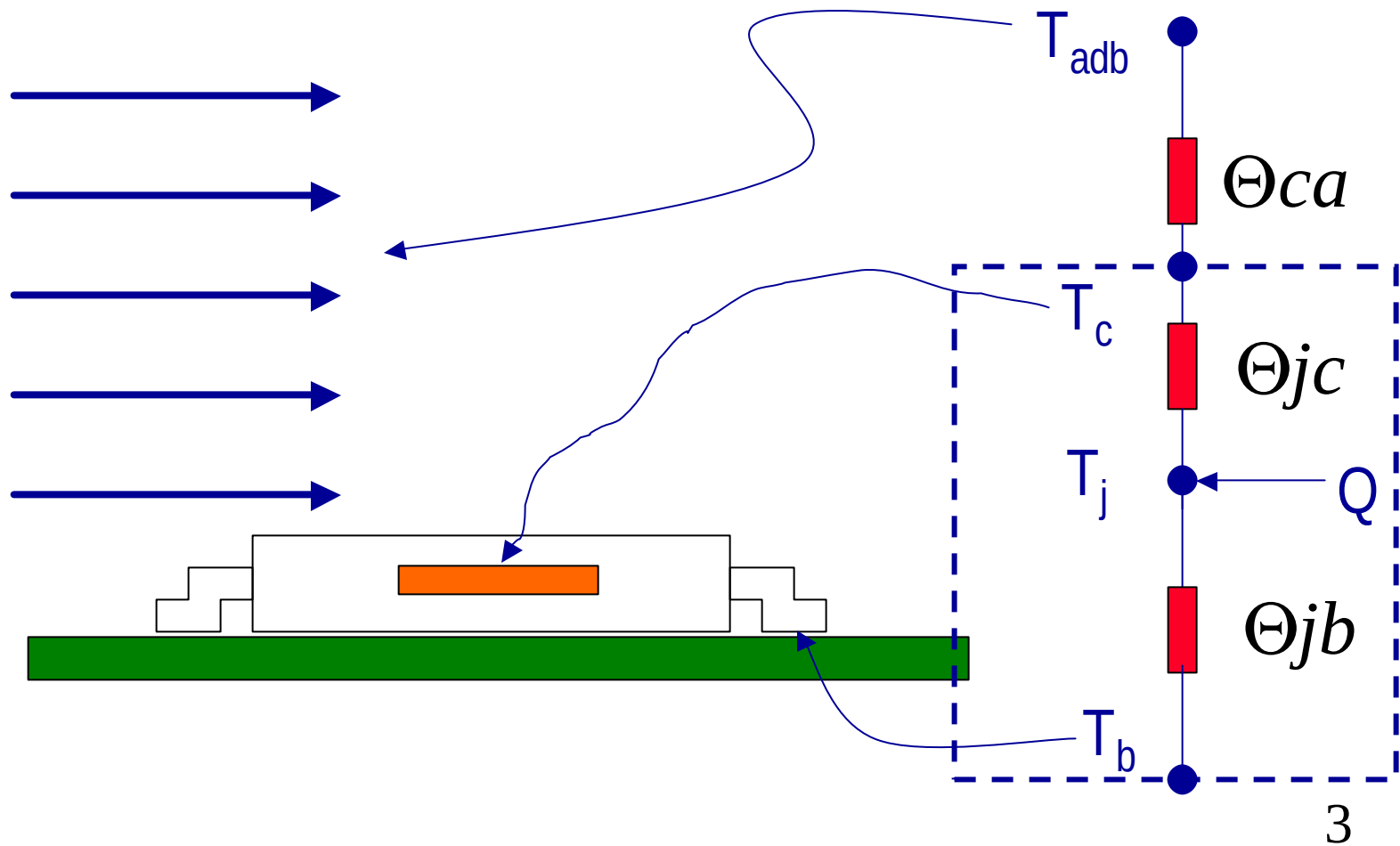


OUTLINE

- I. Some discussion of trends
- II. Air cooling of electronics on PCBs
- III. Natural convection cooling
 - a. In enclosures
 - b. In vertical PCB channels
- IV. Forced convection cooling
 - a. From flush discrete heat sources
 - b. From more realistic three-dimensional sources
- V. Enhanced heat transfer from extended surfaces
 - a. In parallel flow
 - b. In impinging flow
- VI. Future prospects

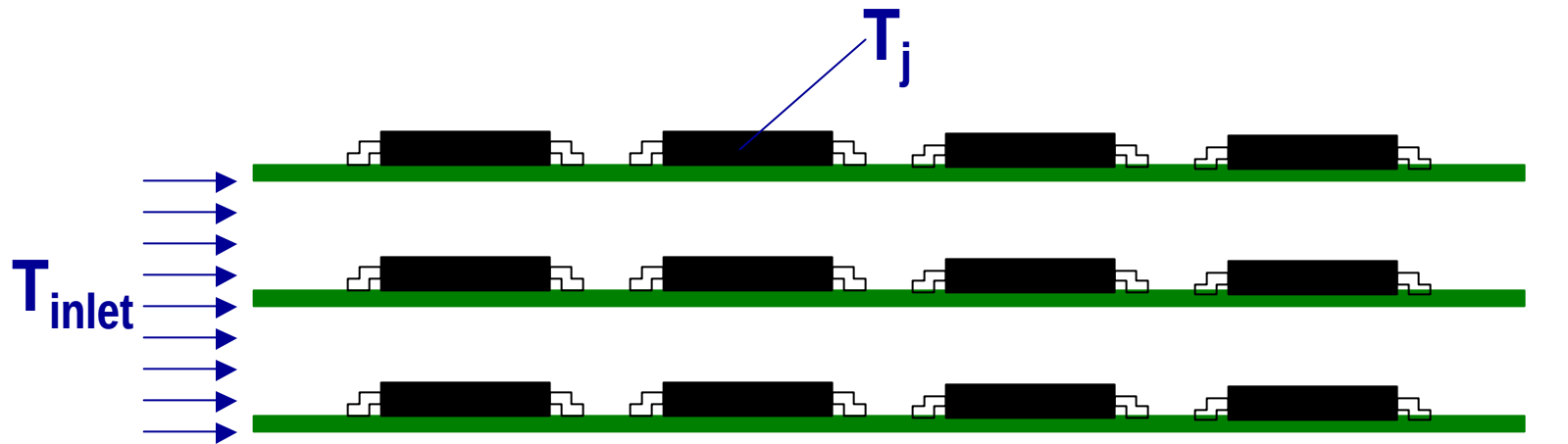


A “Package-centric” View of the World





A “PCB-centric” view of the world: a Hierarchical Problem



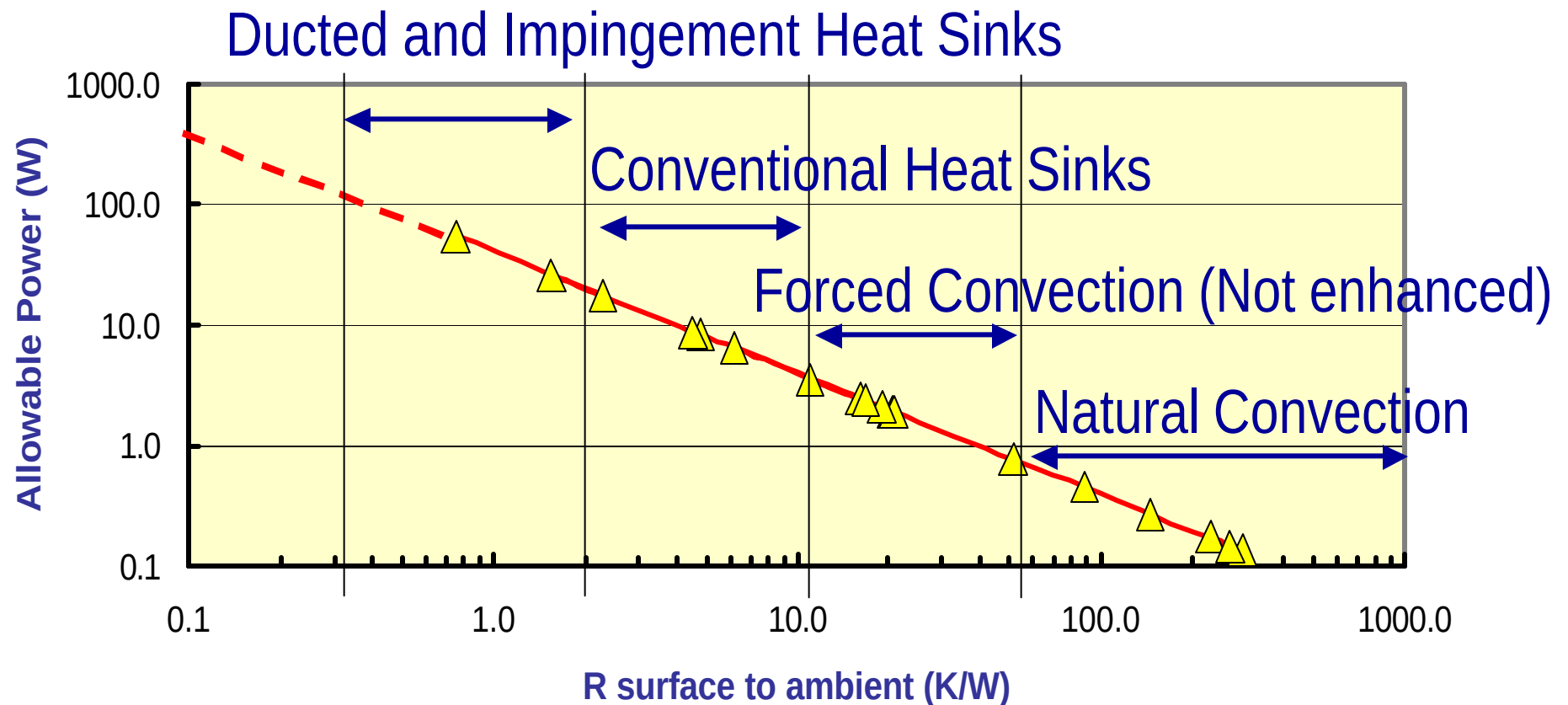
$$T_j = \underbrace{(T_j - T_{case})}_{\text{Internal}} + \underbrace{(T_{case} - T_{adb})}_{\text{Film}} + \underbrace{(T_{adb} - T_{mean})}_{\text{Wake}} + \underbrace{(T_{mean} - T_{inlet})}_{\text{Thermodynamic}} + T_{inlet}$$

Convective + Conductive



Achievable Power Consumption for Various Cooling Modes

Nominal $T_s - T_a = 40^\circ\text{C}$ Planform area = $2.5 \times 2.5 \text{ cm}^2$



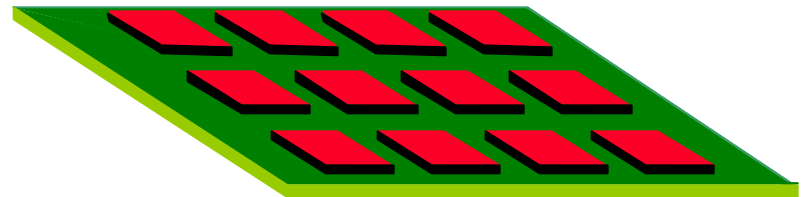
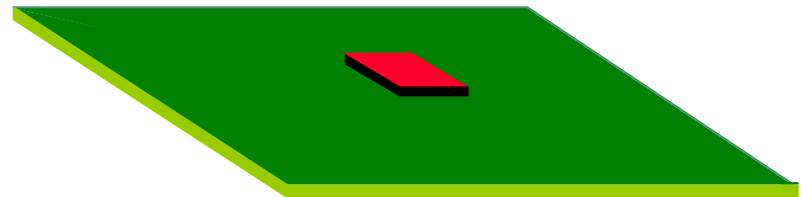
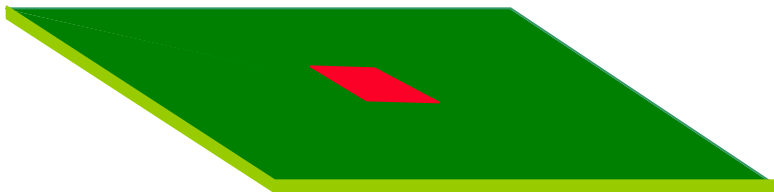
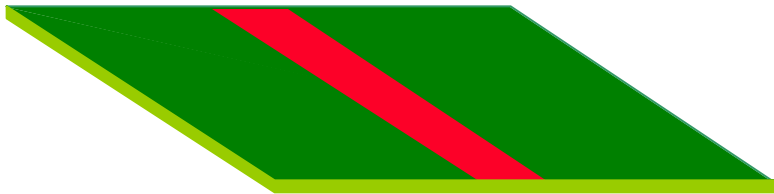


Achievable Power Consumption for Various Cooling Modes

Study	Year	Geometry	Velocity (m/s)	had (W/m ² -K)	Aplanform (cm ²)	R (K/W)	Q (W)
Ortega and Lall	2001	1.0 inch heat source in horizontal encl on FR4	natural convection	7	6.3	228.6	0.2
Horton	1981	Grooved Plates (dips)	natural convection	5.5	6.3	290.9	0.1
Horton	1981	Grooved Plates (flat packs)	natural convection	6	6.3	266.7	0.2
Ortega and Moffat	1985	0.5 inch cubes (no opposing plate)	natural convection	8.5	8.1	145.9	0.3
Ortega and Moffat	1985	0.5 inch cubes (first row H/B=1.5)	bifc	14	8.1	88.6	0.5
Ortega and Wirth	1993	Strip heat source (laminar flow)	low velocity	31	6.3	51.6	0.8
Ortega and Wirth	1993	Strip heat source (turbulent flow)	high velocity	78	6.3	20.5	2.0
Lehmann and Wirtz	1984	Ribs in crossflow	ReH1=12000	36	25.0	11.1	3.6
Wirtz and Dykshoorn	1984	Sparse Flatpacks	Rechannel=3700	48	12.9	16.1	2.5
Arvizu and Moffat	1981	Sparse Cubes	1.5 m/s	60	8.1	20.7	1.9
Arvizu and Moffat	1981	Dense Cubes	1.5 m/s	65	8.1	19.1	2.1
Arvizu and Moffat	1981	Dense Cubes	5 m/s	200	8.1	6.2	6.5
Ortega and Singer	1997	Impinging Slot Jet	low velocity	95	6.3	16.8	2.4
Ortega and Singer	1997	Impinging Slot Jet	high velocity	330	6.3	4.8	8.3
Ortega and Urdaneta	2001	Pin Fin Heat Sink - Side In Side Out	1.5 m/s		6.3	4.5	8.9
Ortega and Urdaneta	2001	Pin Fin Heat Sink - Side In Side Out	6.6		6.3	1.6	25.6
Ortega and Issa	2001	Pin Fin Heat Sink - Top In Side Out	1.94		6.3	2.3	17.4
Ortega and Issa	2002	Pin Fin Heat Sink - Top In Side Out	20		6.3	0.8	53.3



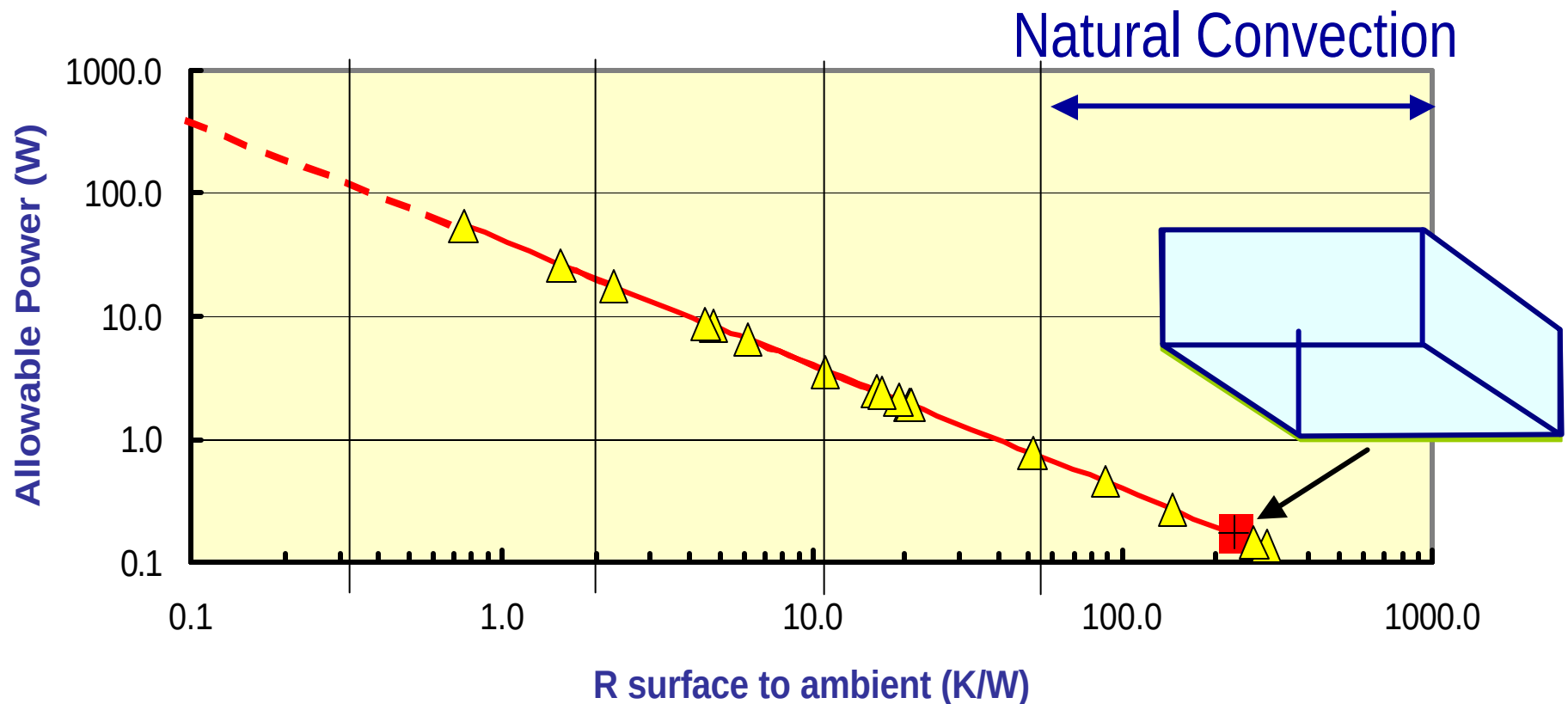
Paradigms for PCB Cooling





Natural Convection in a Horizontal Enclosure

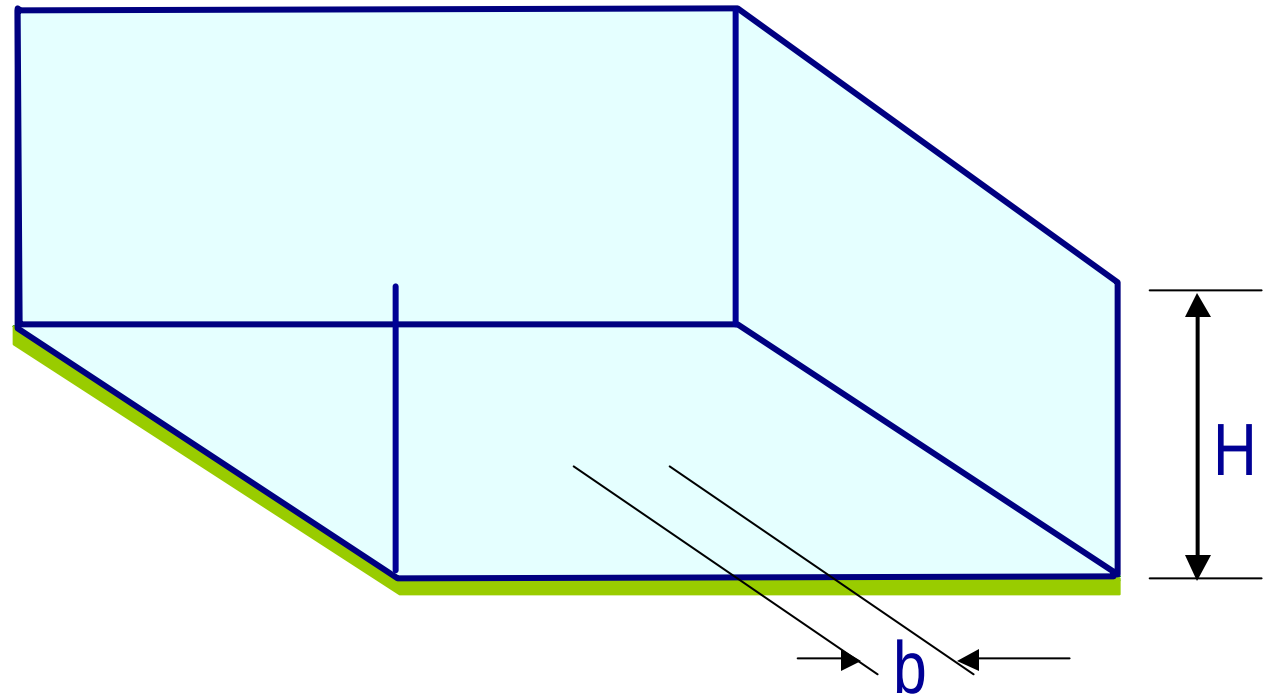
Nominal $T_s - T_a = 40\text{ C}$ Planform area = $2.5 \times 2.5\text{ cm}^2$





Natural Convection in a Horizontal Enclosure

- ◆ What is the effect of enclosure height
- ◆ How does board conduction affect h
- ◆ Code Validation experiment

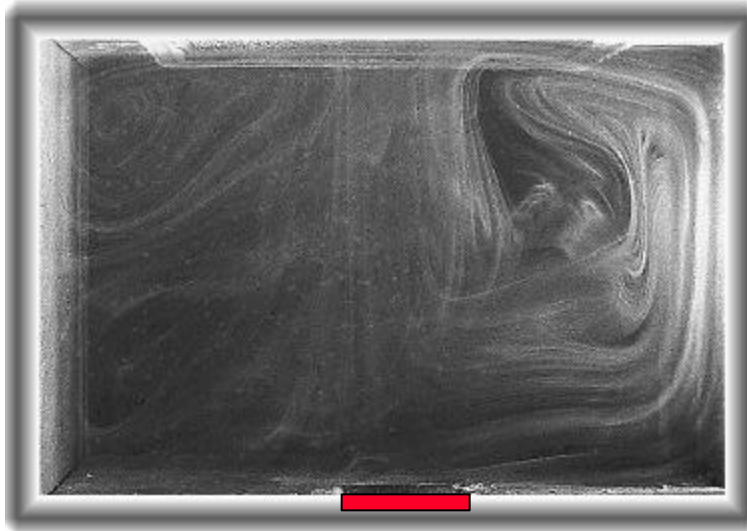


Lall and Ortega (1996, 2001)

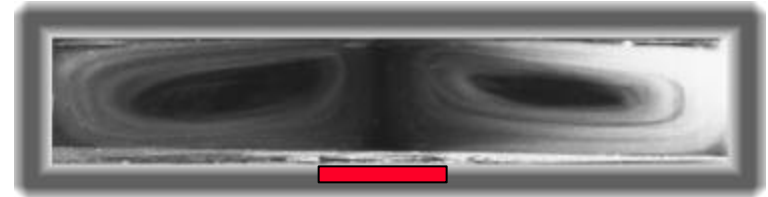


Flow Visualization in Enclosure with Adiabatic Board: Various H/b

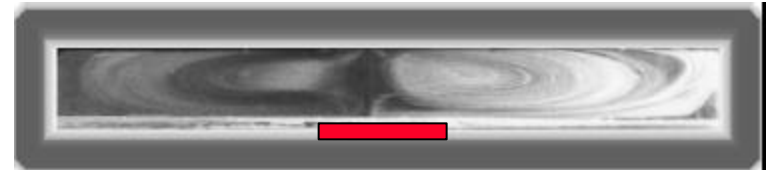
4



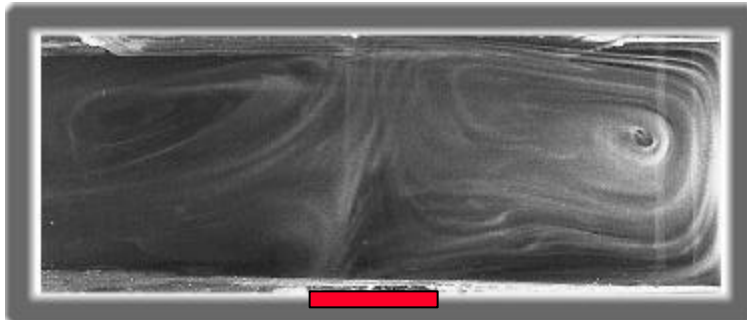
1



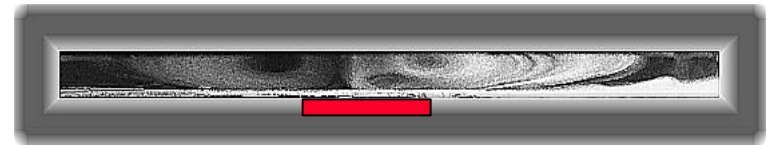
1/2



2



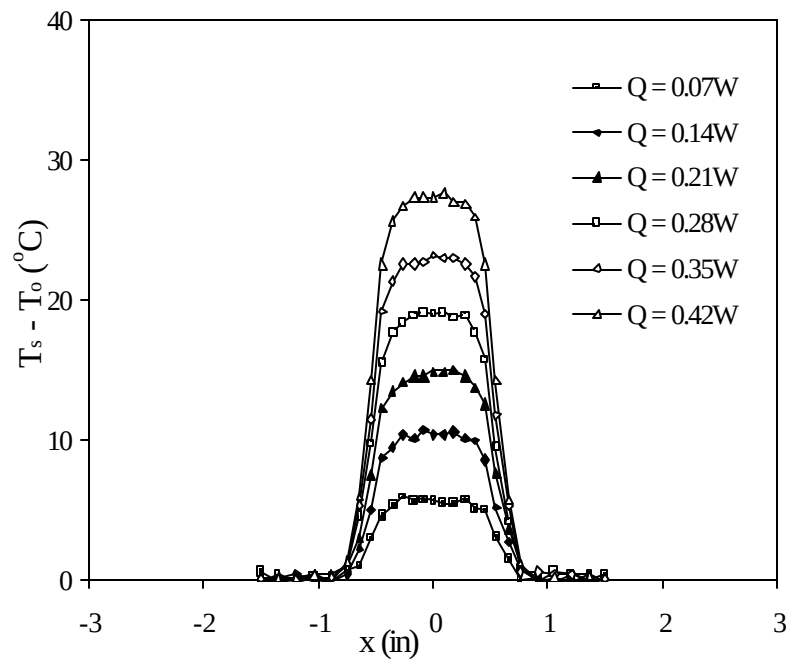
1/4



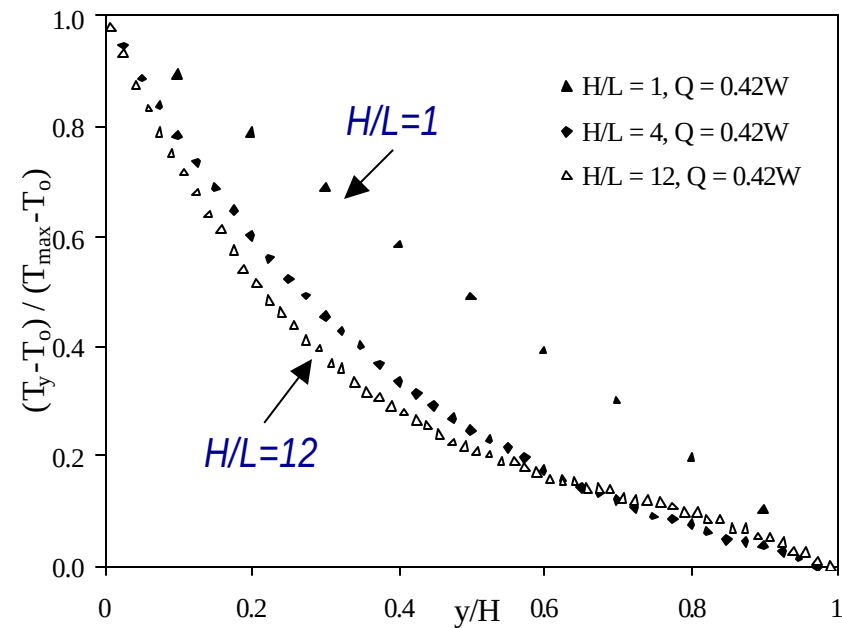


Temperatures: Adiabatic Board

*Plate Surface Temperature:
 $H/L=1$*



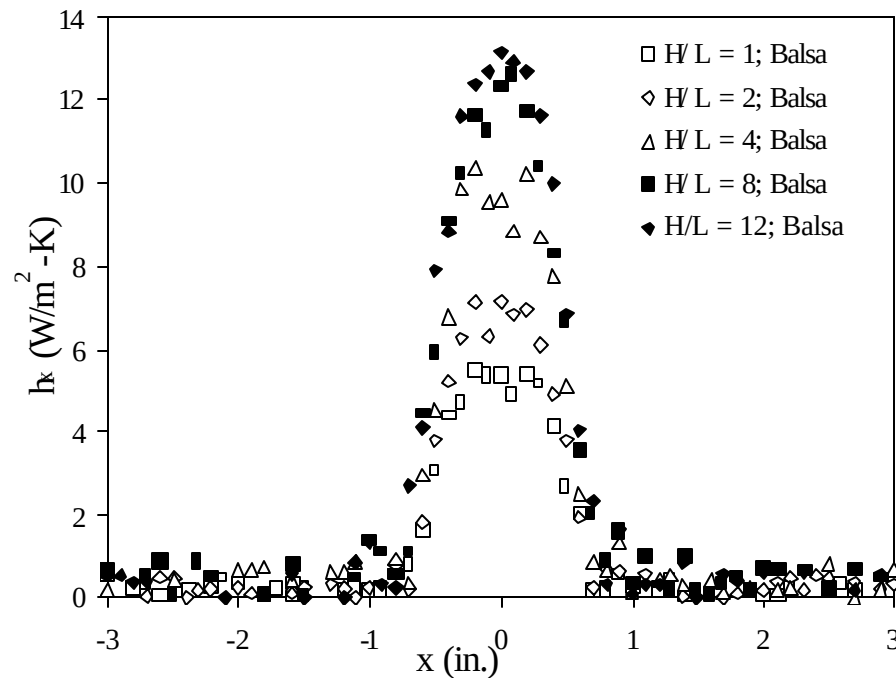
*Air Temperature above
Source Centerline*



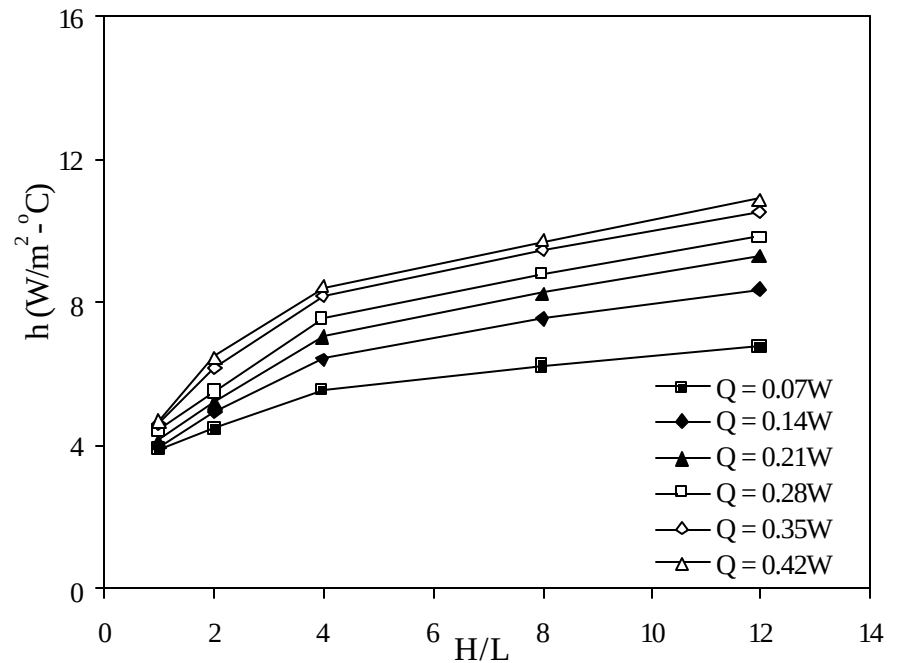


Heat Transfer Coefficient: Adiabatic Board

*Heat Transfer Coefficient
0.42 W Various H/L*

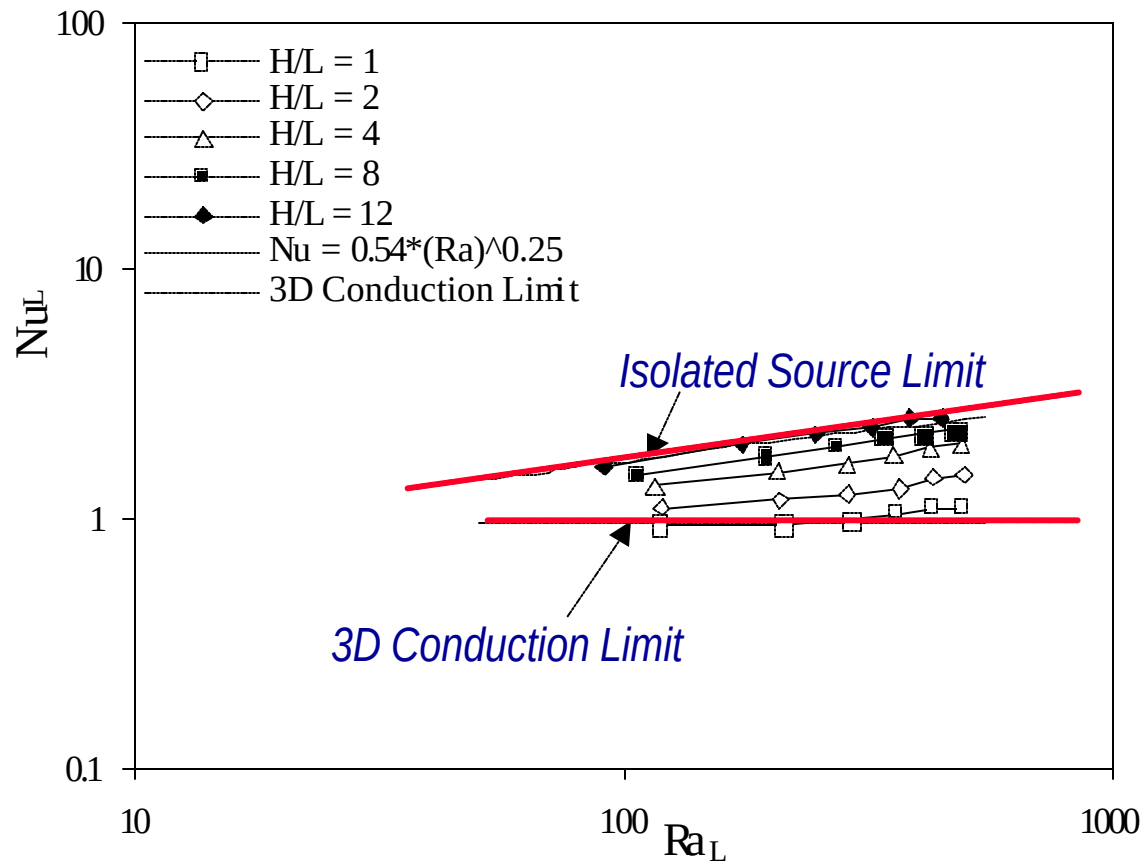


*Source Average
Heat Transfer Coefficient*





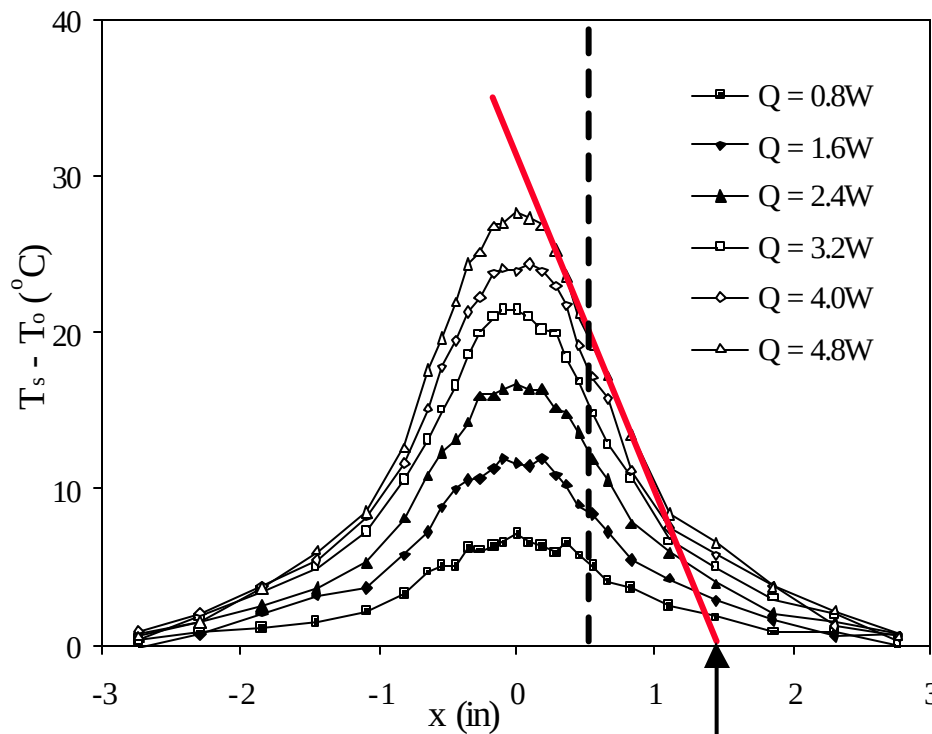
Source Average Nusselt Number: Adiabatic Board



- Source Average Nusselt Number is bounded by a Conduction Limit and an Isolated Source Limit
- Conduction Limit for $H/L < 1$
- The Isolated Source Limit well described by traditional horizontal flat plate correlations



The Thermal Footprint Radius: A Conjugate Length Scale



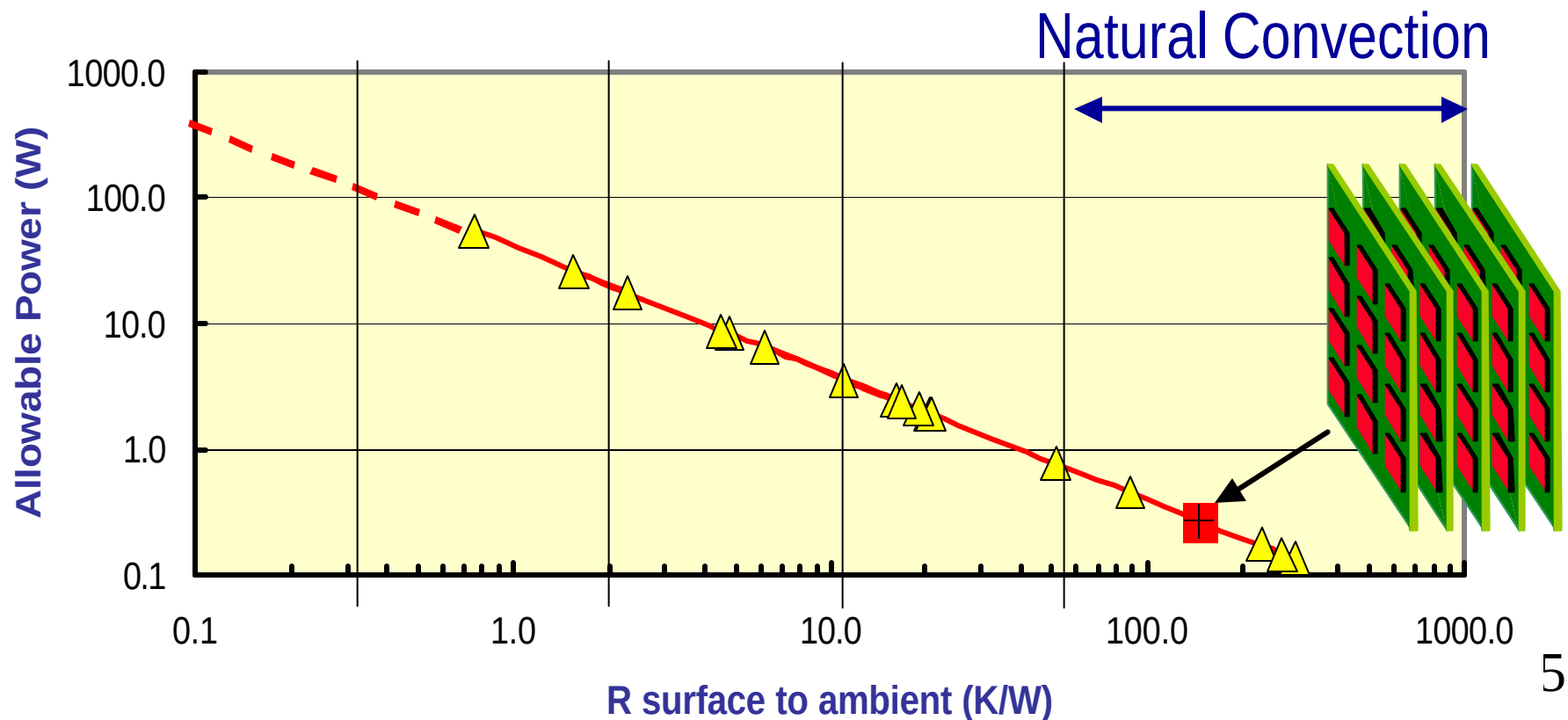
- Thermal Footprint radius depends on source length scale and board conduction properties
- The effective size of the source, including heat spreading effects, are captured by the footprint radius

Thermal Footprint Radius r_f



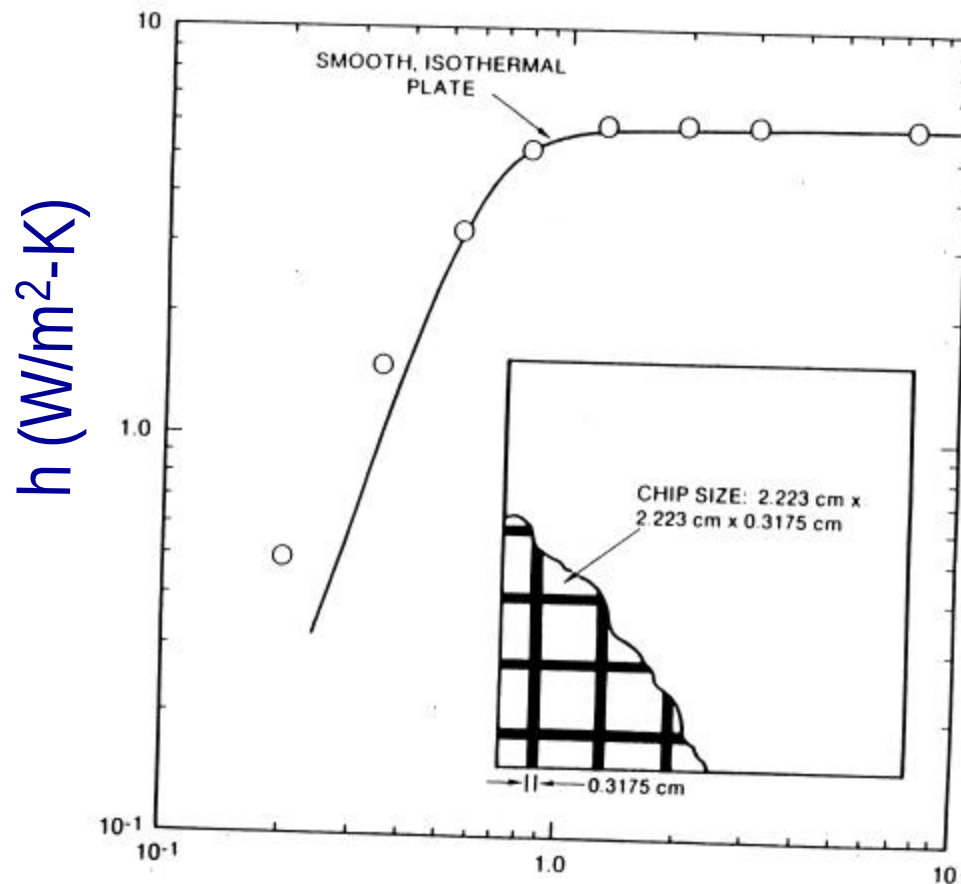
Natural Convection Cooling in Vertical Card Stack

Nominal $T_s - T_a = 40\text{ C}$ Planform area = $2.5 \times 2.5\text{ cm}^2$

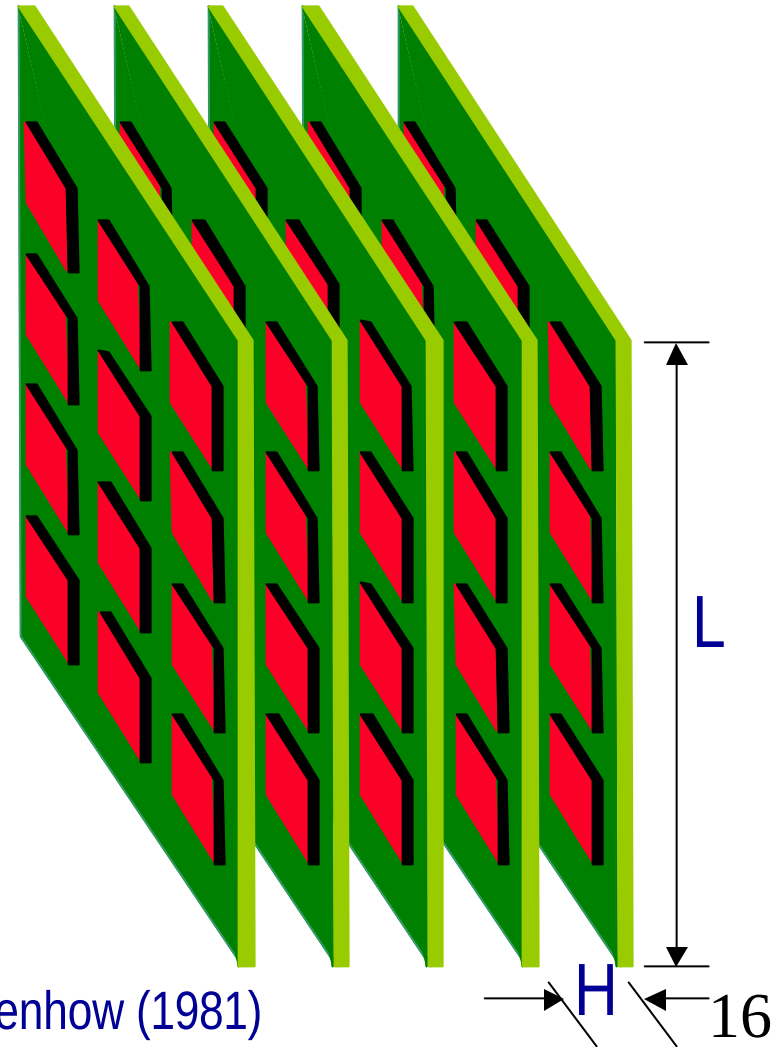




Natural Convection Cooling in Vertical Card Stack

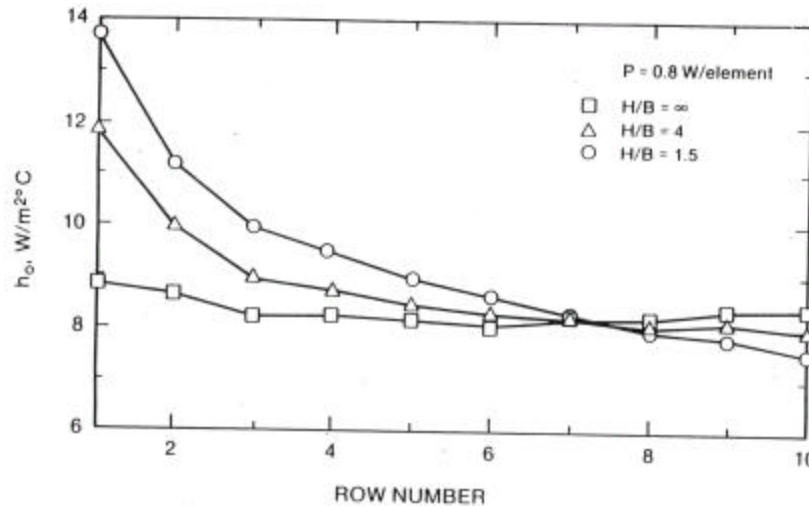


H (cm) Horton and Rosenhow (1981)





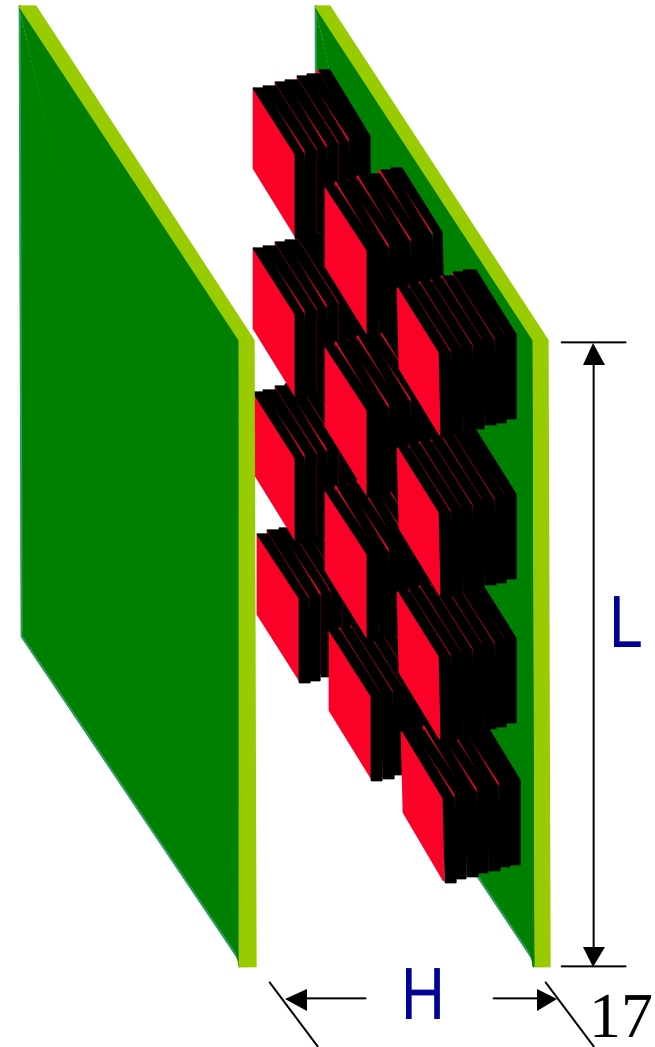
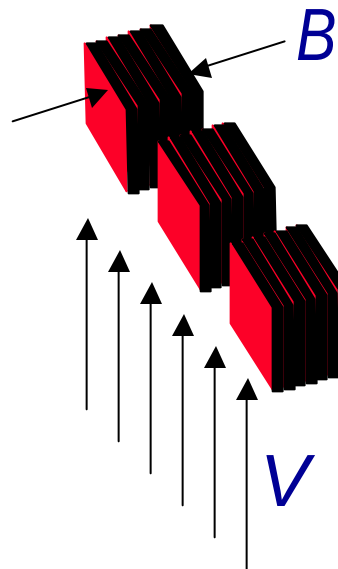
Buoyancy Induced Forced Convection in Array of Cubes



$$\text{Local Reynolds Number} = \frac{VB}{n}$$

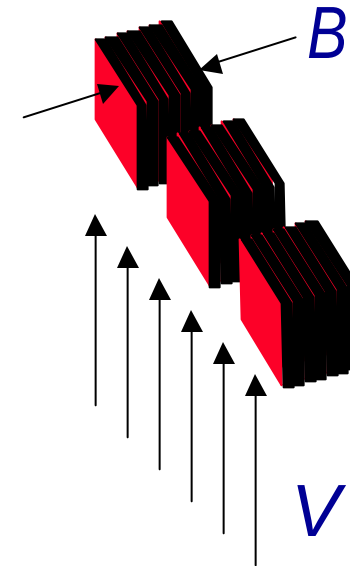
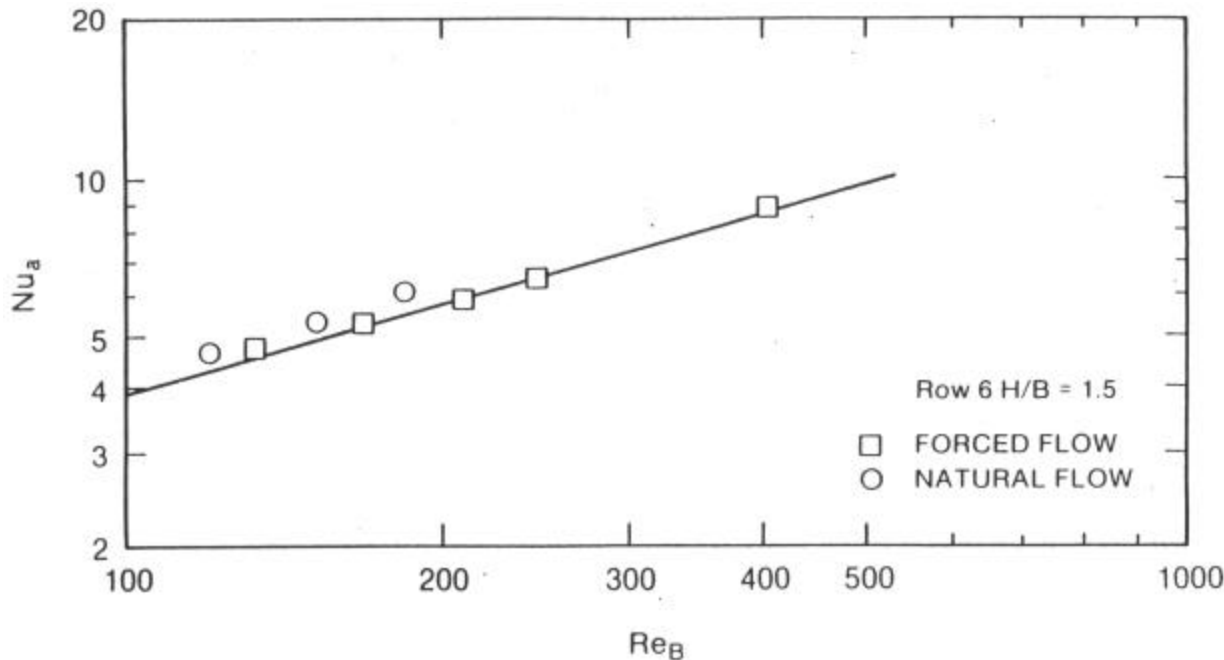
$$\text{Local Grashof Number} = \frac{gb(T - T_{ad})B^3}{n^2}$$

Ortega and Moffat (1985, 1986)





Buoyancy Induced Forced Convection in Array of Cubes



- ◆ Global chimney flow causes the local heat transfer to be in “forced convection”
- ◆ We call it Buoyancy Induced Forced Convection
- ◆ Knowledge from Forced Convection situation applies directly

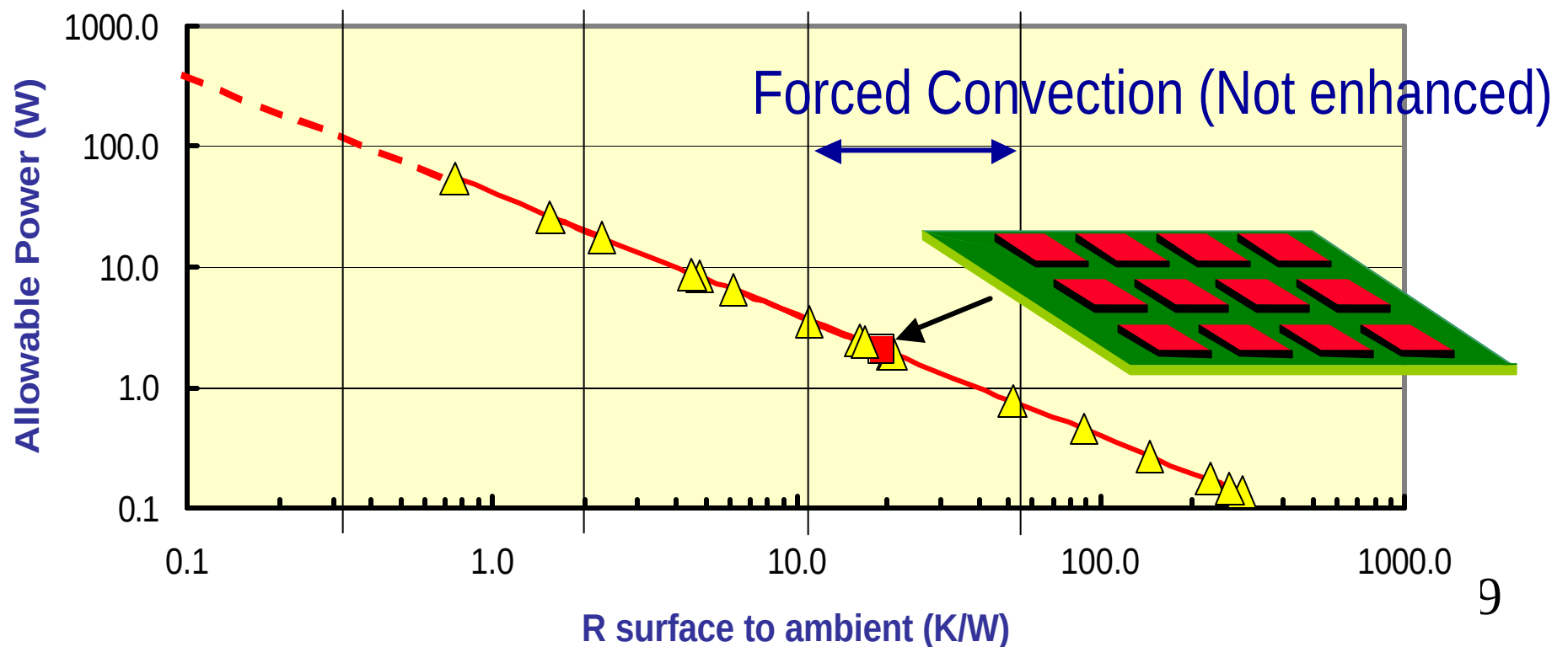
$$\text{Local Reynolds Number} = \frac{VB}{n}$$

$$\text{Local Grashof Number} = \frac{gb(\tau - \tau_{ad})B^3}{n^2}$$



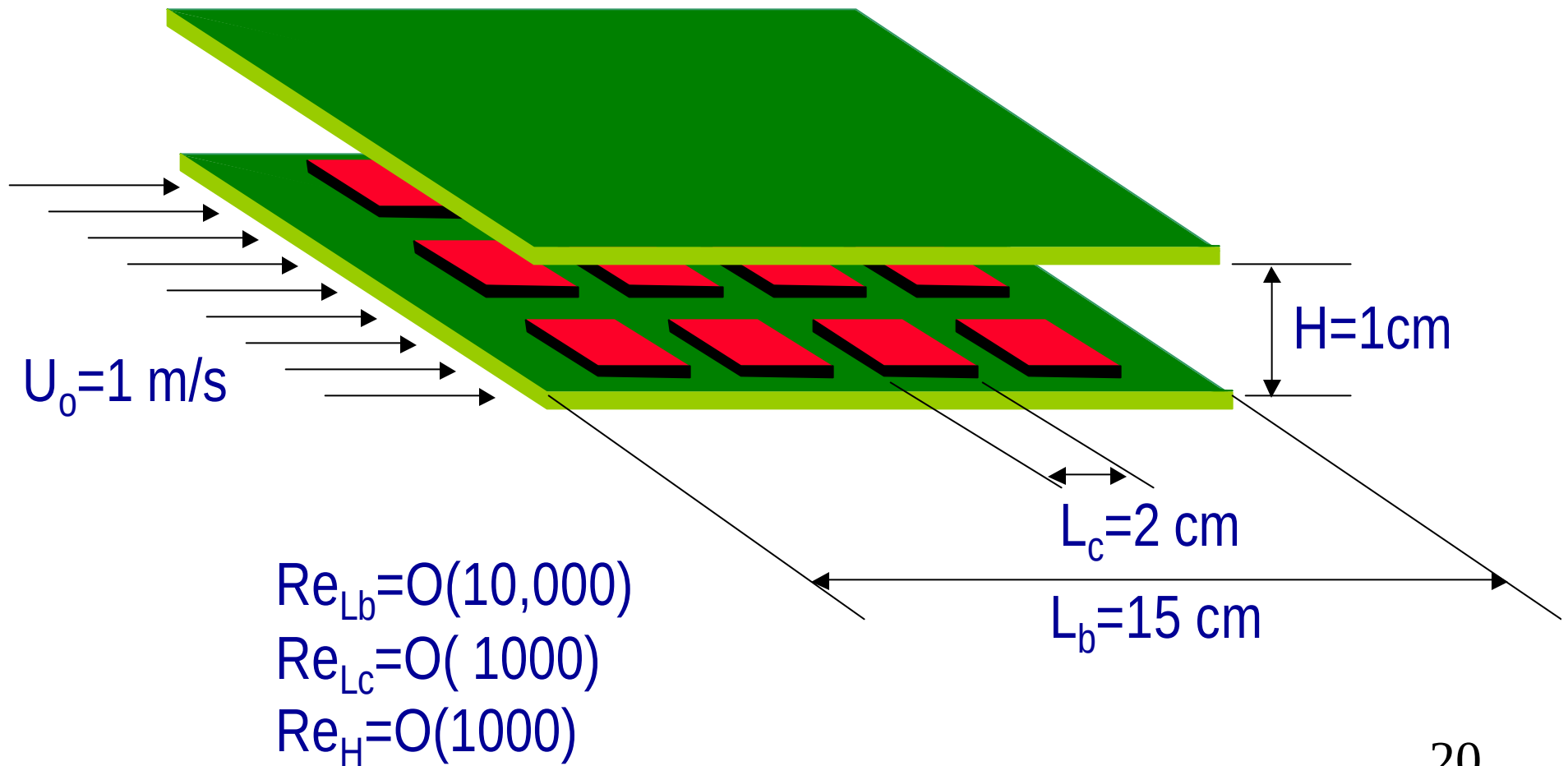
Forced Convection: Array of cubes on an insulated board

Nominal $T_s - T_a = 40\text{ C}$ Planform area = $2.5 \times 2.5\text{ cm}^2$



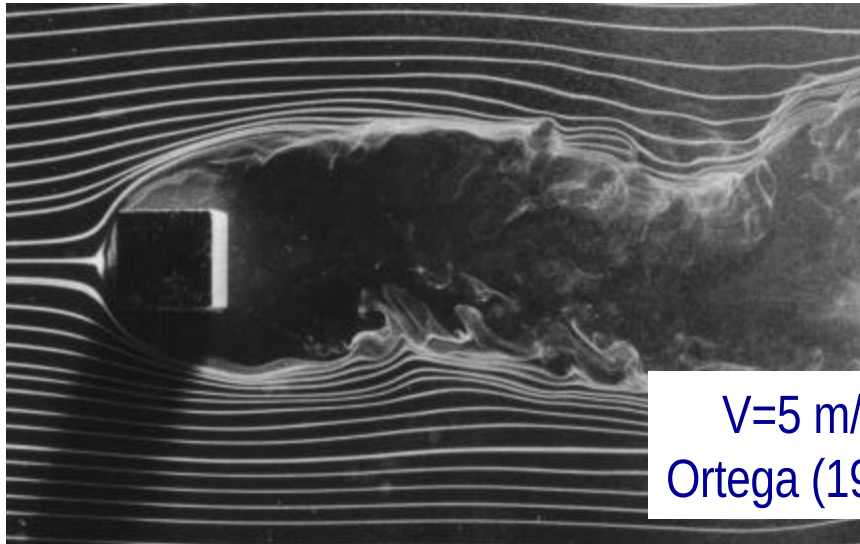


Forced Convection: Array of cubes on an insulated board

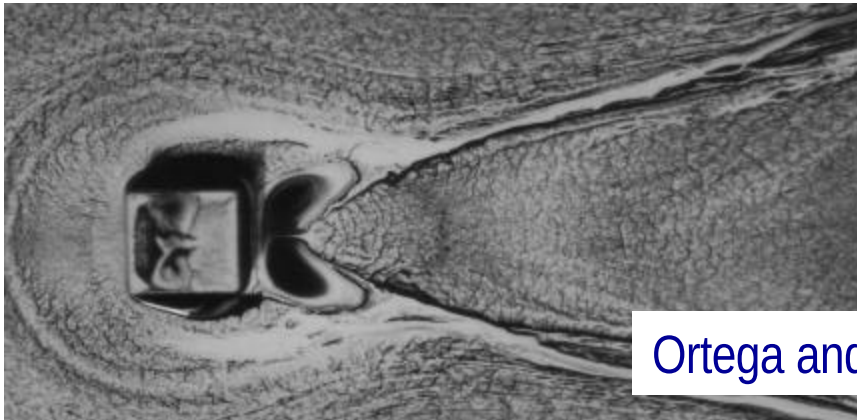




Physics of flow about a cube immersed in an air boundary layer



$V=5$ m/s
Ortega (1984)

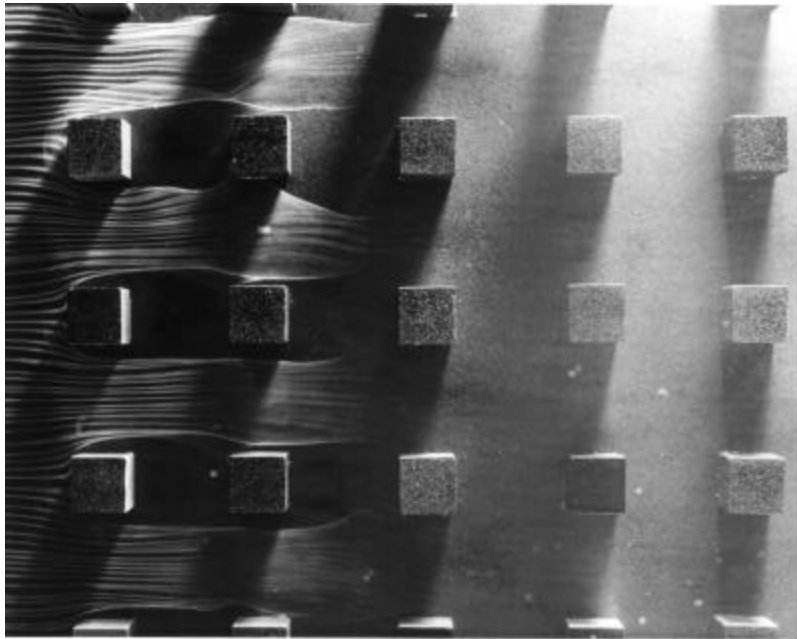


Ortega and Meinders (1999)

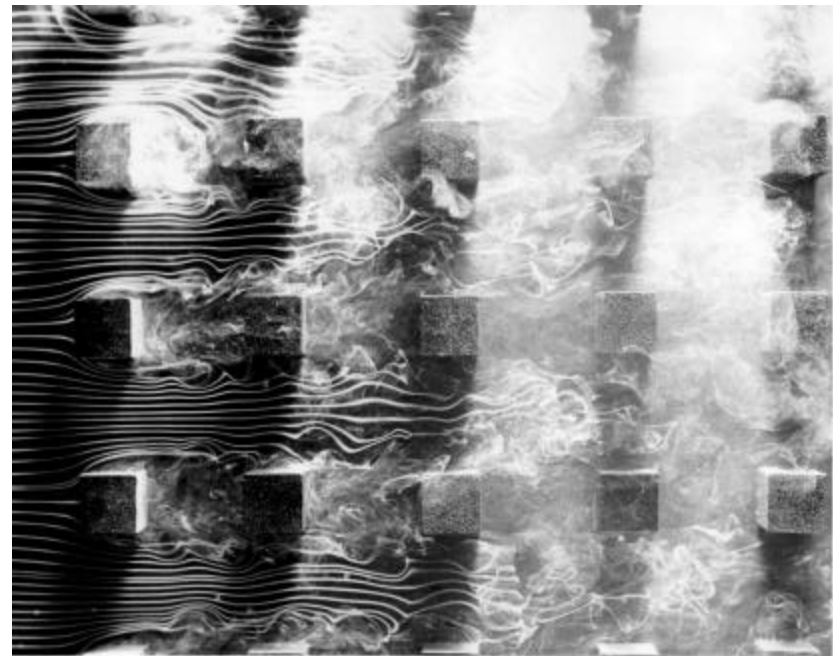
- ♦ Flow typically is low Reynolds number, unsteady, laminar-turbulent transitional
- ♦ Complex three-dimensionality
- ♦ Heat transfer scales on local parameters, including local velocity, local free-stream turbulence, and local adiabatic temperature



Flow about an array of cubes immersed in an air boundary layer



$V=0.8$ m/s $H/B = 4$ $S/B = 3$
Laminar



$V=5$ m/s $H/B = 4$ $S/B = 3$
Transitional



Dealing with discontinuous surface heating

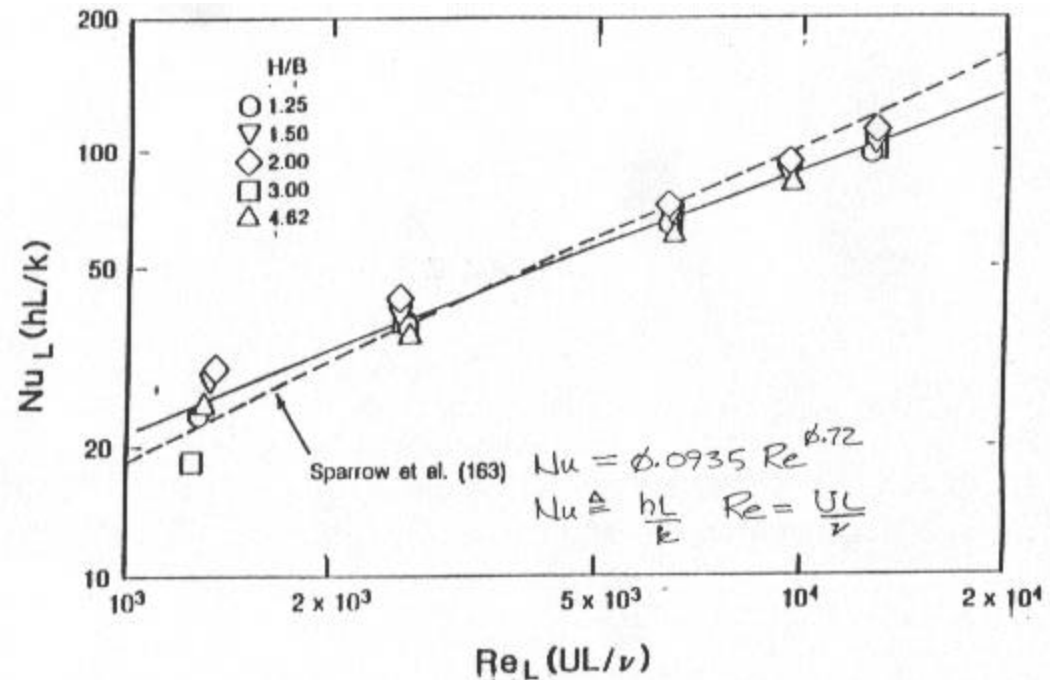
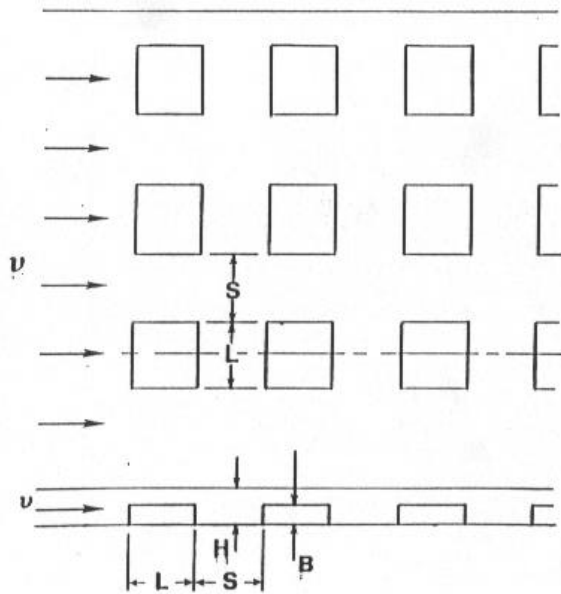
$$T_j = (T_j - T_{\text{case}}) + \underbrace{(T_{\text{case}} - T_{\text{adb}})}_{\frac{q''}{h_{\text{adb}} A_{\text{surface}}}} + \underbrace{(T_{\text{adb}} - T_{\text{mean}})}_{\text{thermal wake}} + (T_{\text{mean}} - T_{\text{inlet}}) + T_{\text{inlet}}$$

Predicting local temperature rise requires that we know the local adiabatic heat transfer coefficient

Predicting component interactions requires that we know the thermal wake behavior



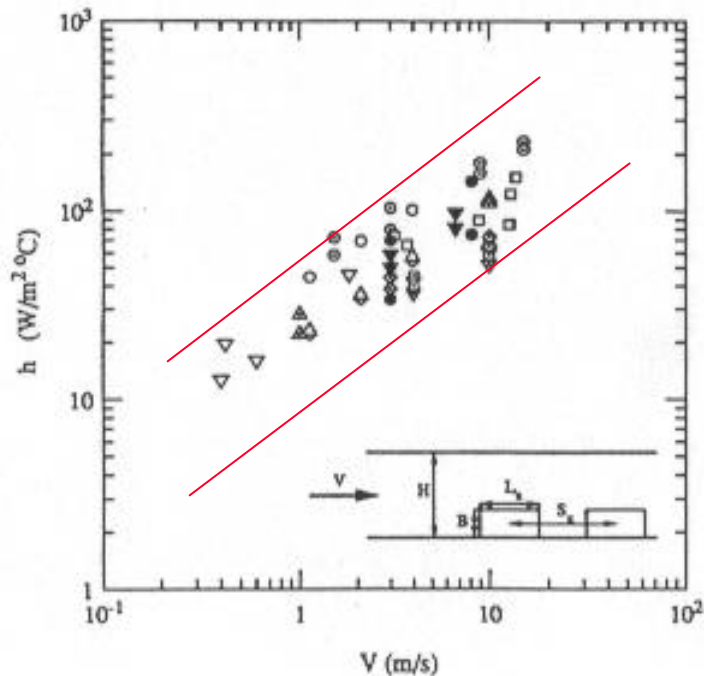
Empirically Measured adiabatic h for array of Flat Packages



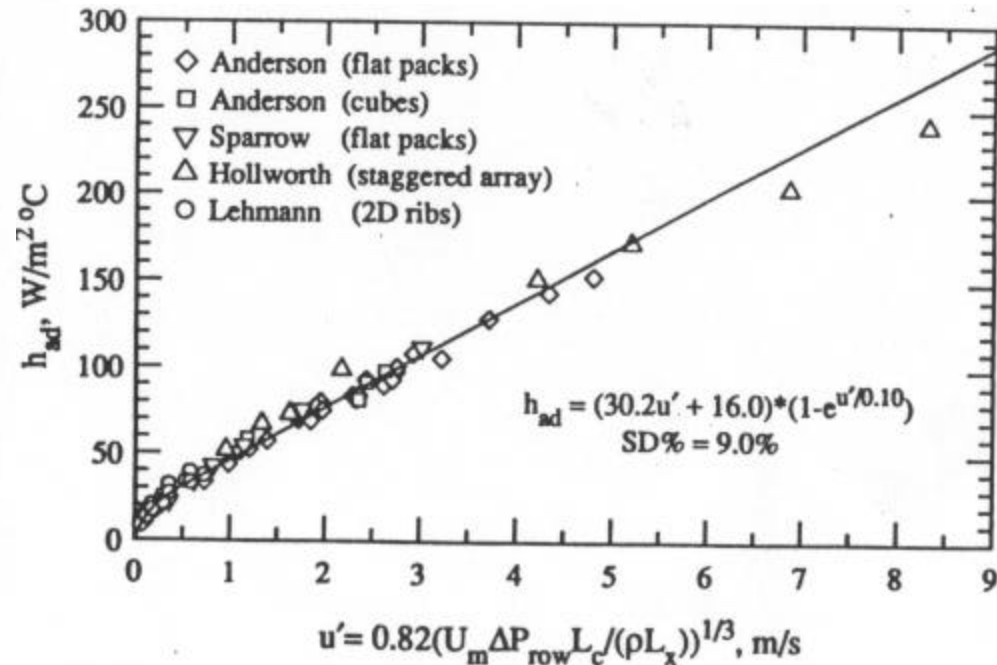
Sparse Flatpacks in Laminar and Transitional Flow
Wirtz and Dykshoorn (1984)



Empirically Measured adiabatic h for wide range of geometries



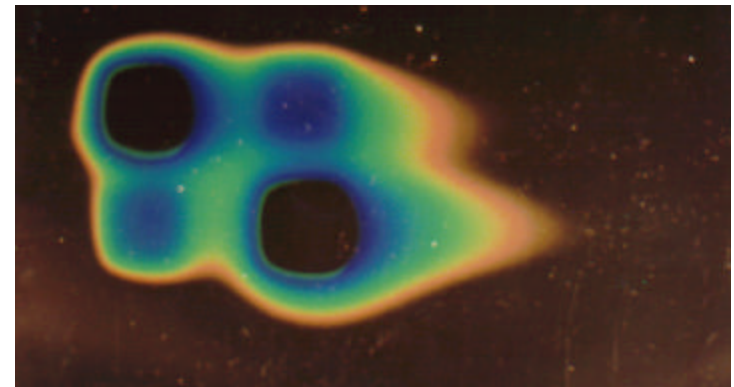
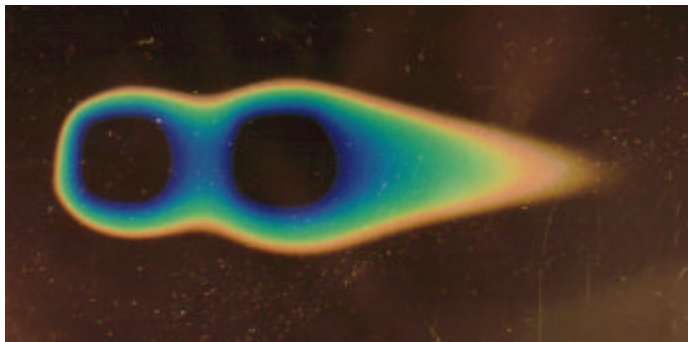
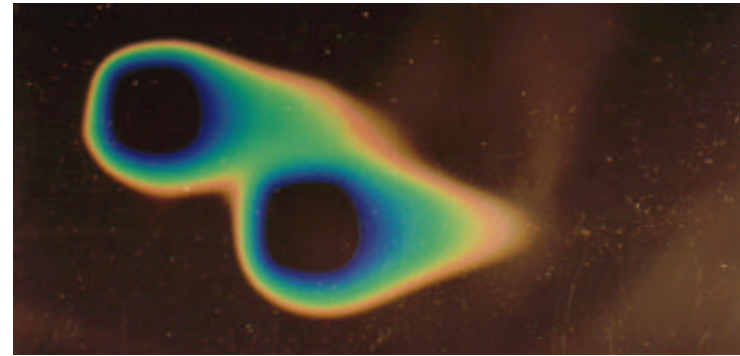
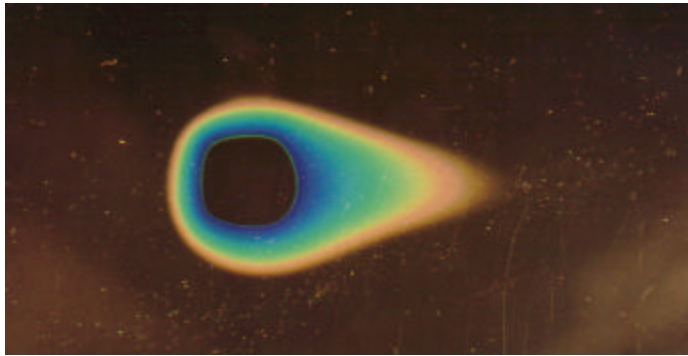
There are too many geometric lengthscale to allow successful development of traditional empirical correlations



h_{ad} scales well on local inferred maximum turbulent intensity, which can be scaled with pressure drop

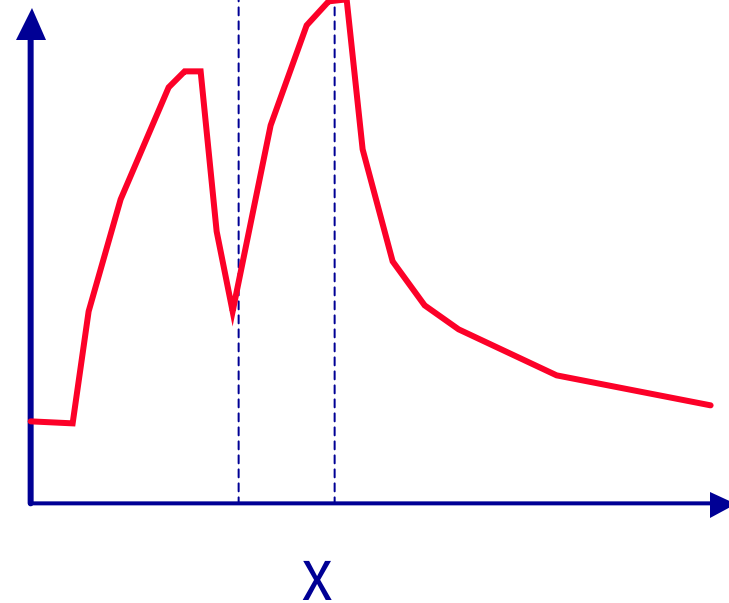
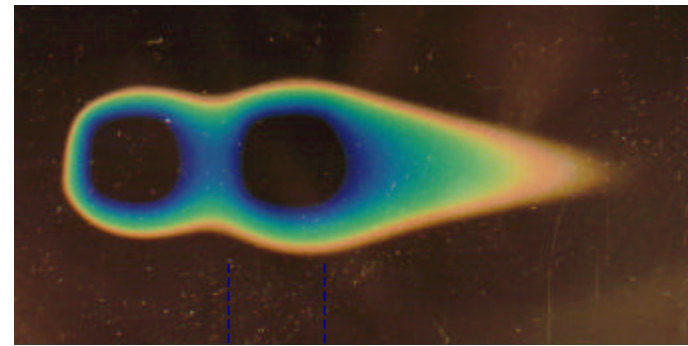
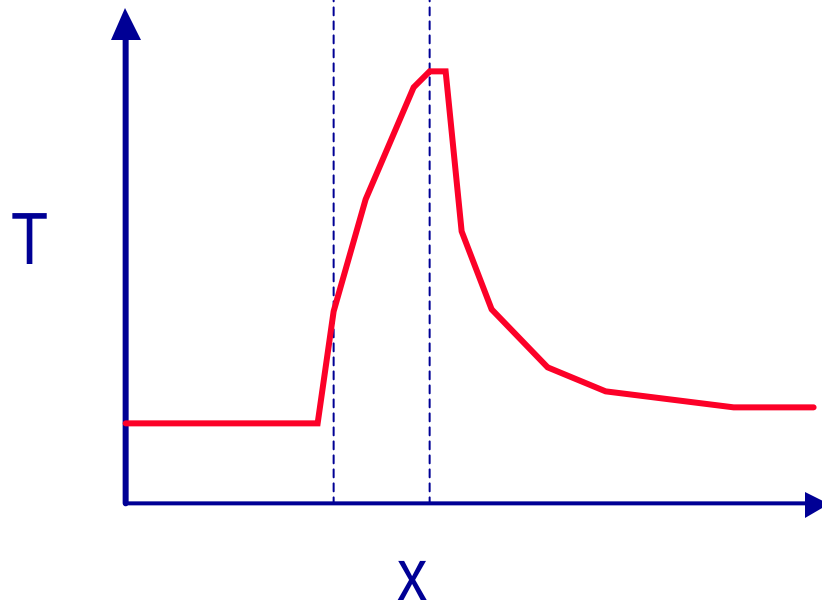
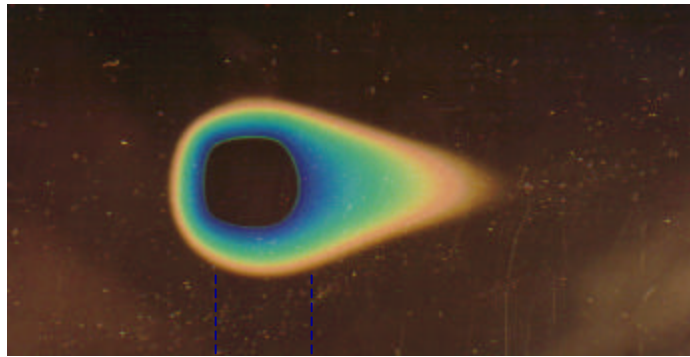


Component Interactions





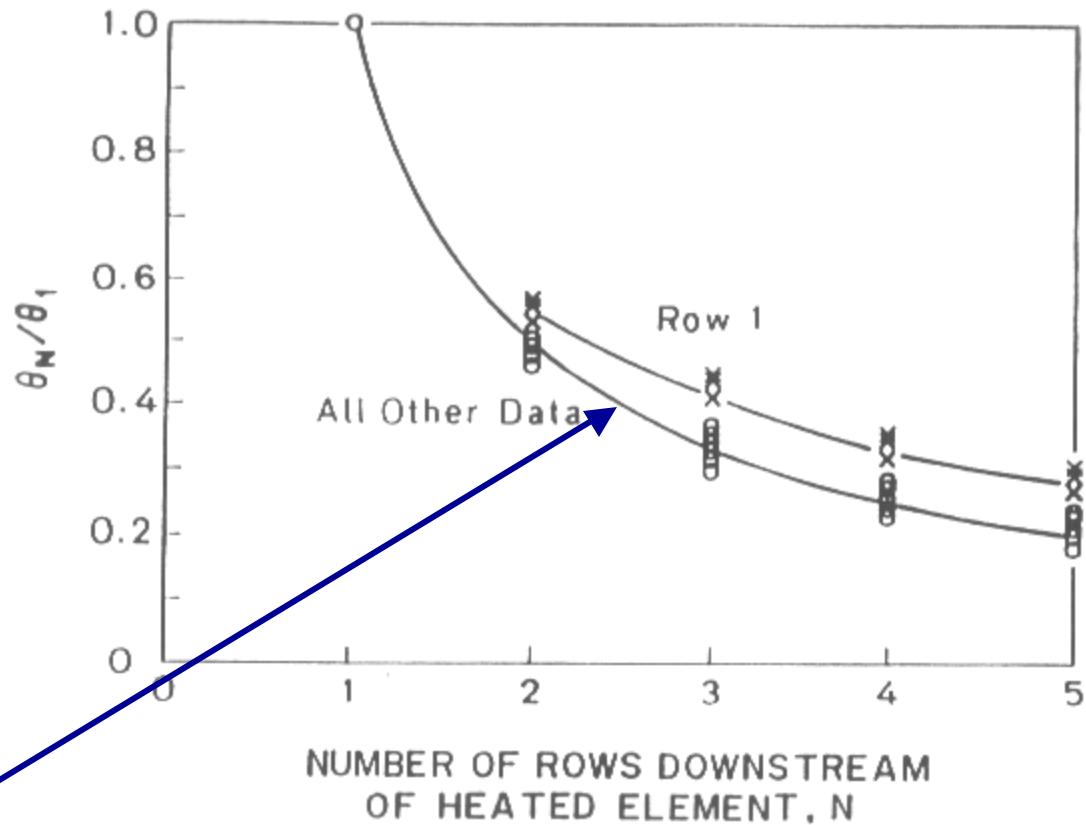
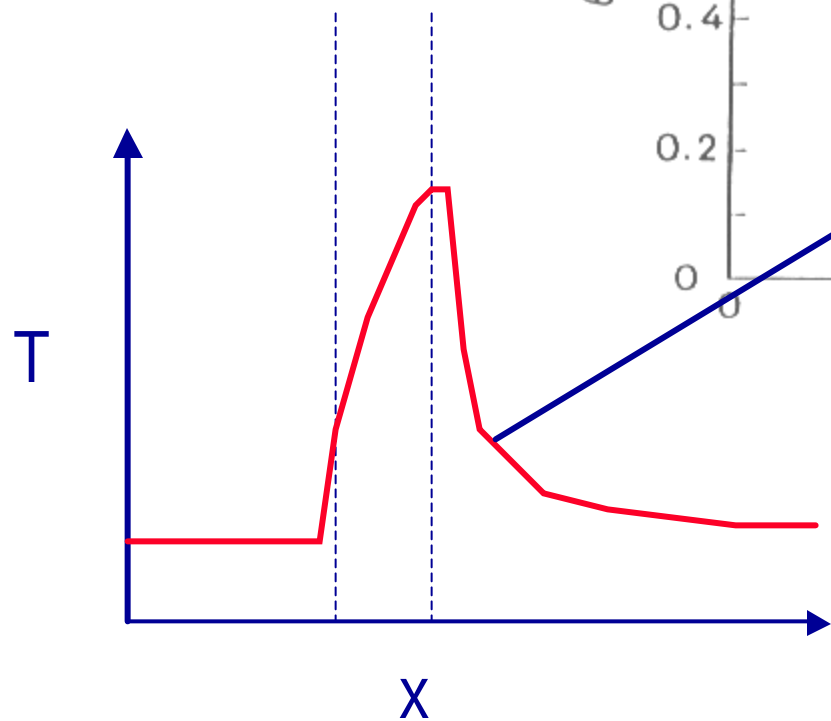
Component Interactions





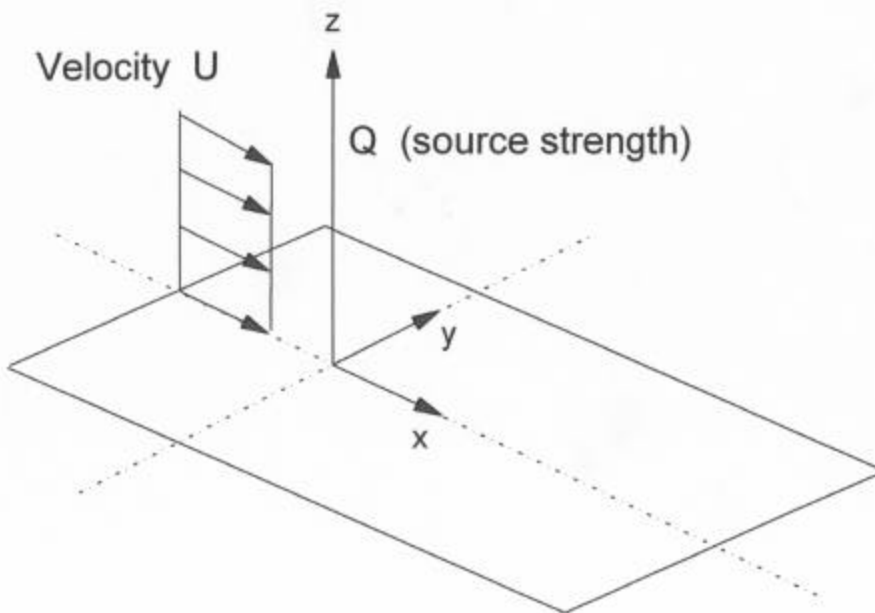
Empirical model of Thermal Wake along its centerline

$$\Theta_N / \Theta_1 = 1 / N$$





Thermal Greens Functions



- ◆ The Thermal Greens function is the solution of the temperature due to a point source of heat
- ◆ It provides the relationship between the surface heating and the thermal response
- ◆ Hence, it embodies the same information as the heat transfer coefficient



Thermal Greens Functions

- ◆ The thermal Greens function for uniform flow is given by

$$\theta = \left\{ \frac{q}{2\pi\rho c \epsilon R} \right\} e^{-\frac{U}{2\epsilon}(R-x)}$$

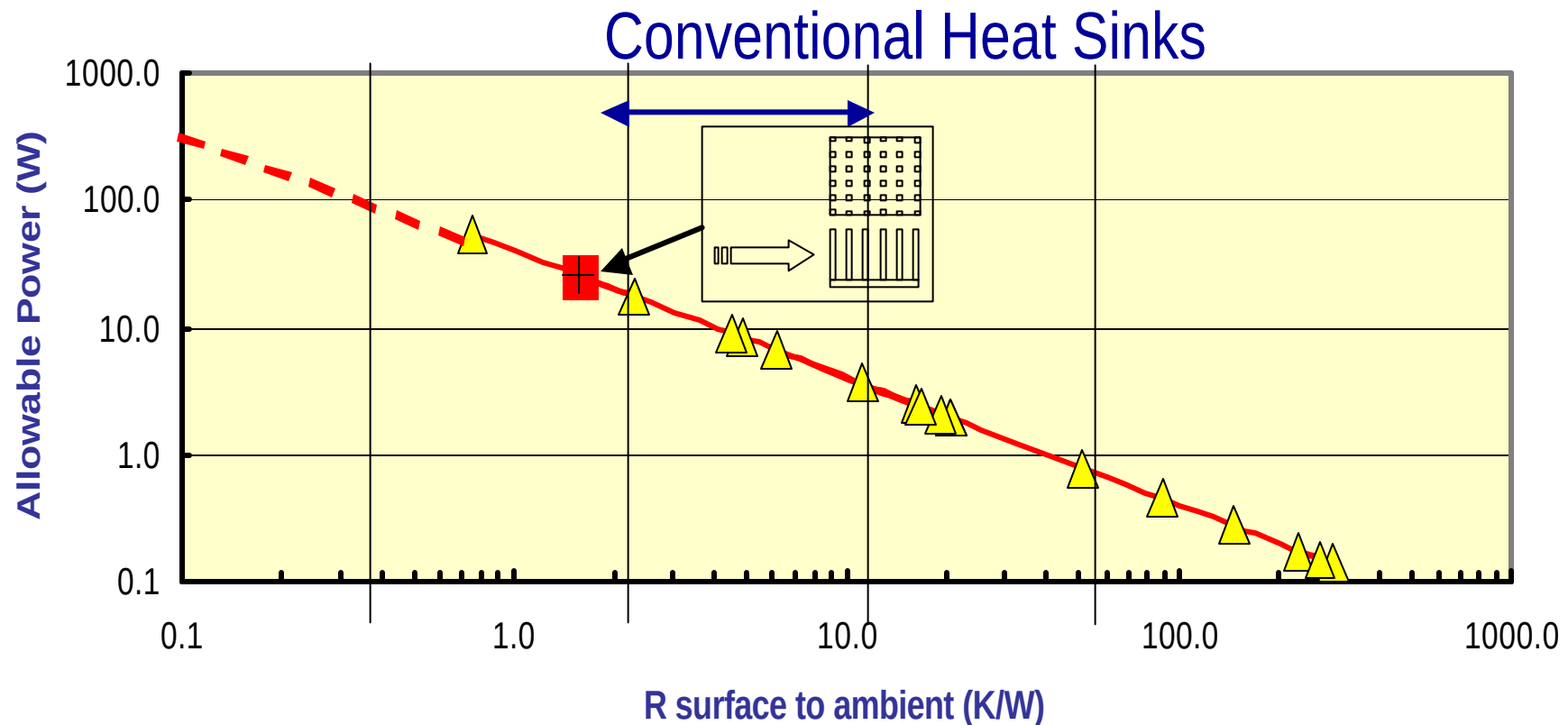
$$R = (x^2 + y^2)^{1/2}$$

- ◆ Note that at the surface, it predicts the 1/x dependence of the thermal wake



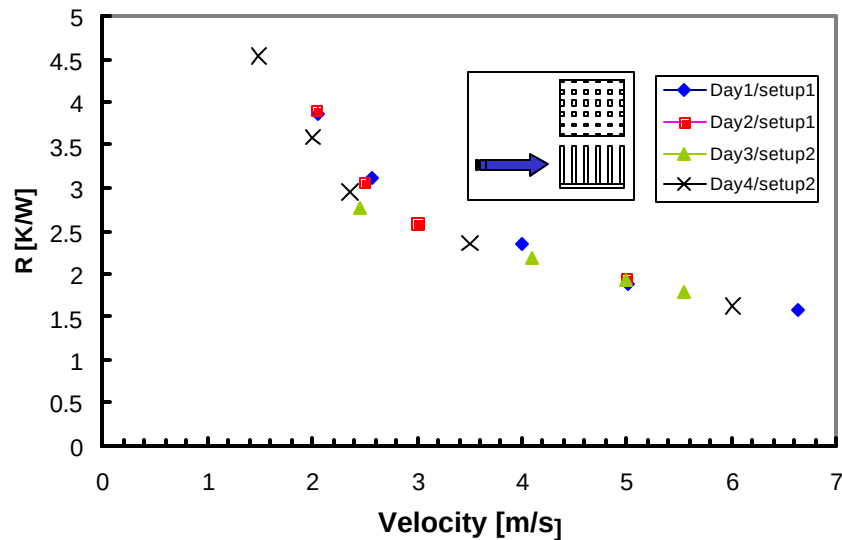
Forced Convection: Pin Fin Heat Sinks in cross flow

Nominal $T_s - T_a = 40^\circ\text{C}$ Planform area = $2.5 \times 2.5 \text{ cm}^2$

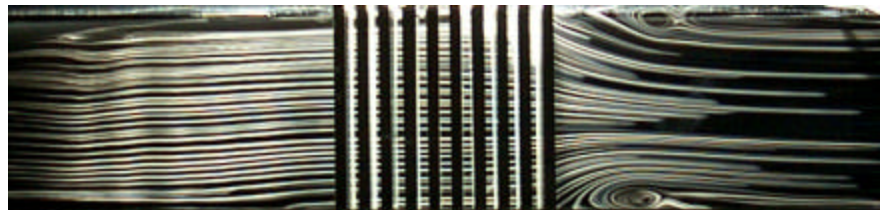




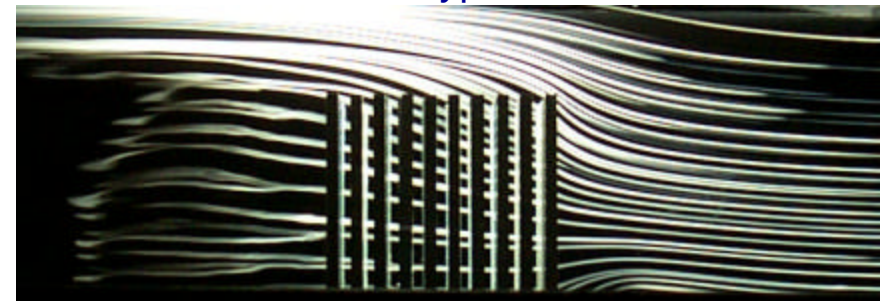
Heat Transfer: Conventional Pin Fin Heat Sinks



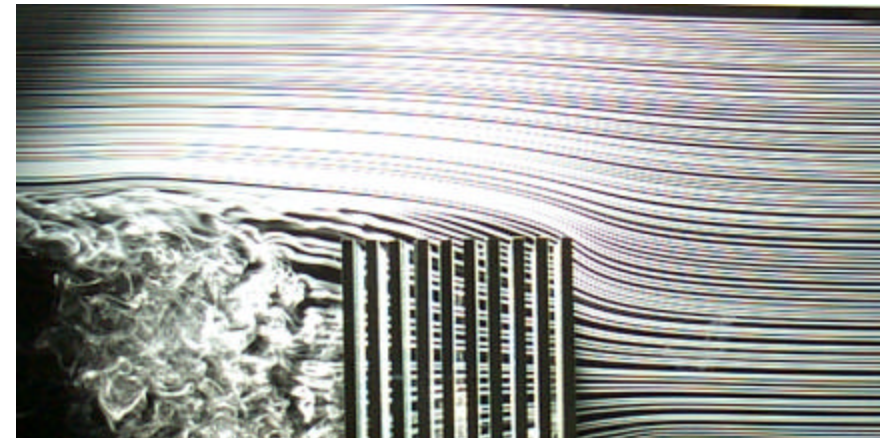
- ◆ Conventional Plate Fin and Pin Fin heat sinks bottom out at R greater than 0.5 K/W even for velocities in excess of 5 m/s
- ◆ Plate Fin heat sinks generally have better thermal performance for given pressure drop



$S/t=2$ Bypass=0



$S/t=2$ Bypass=1/2

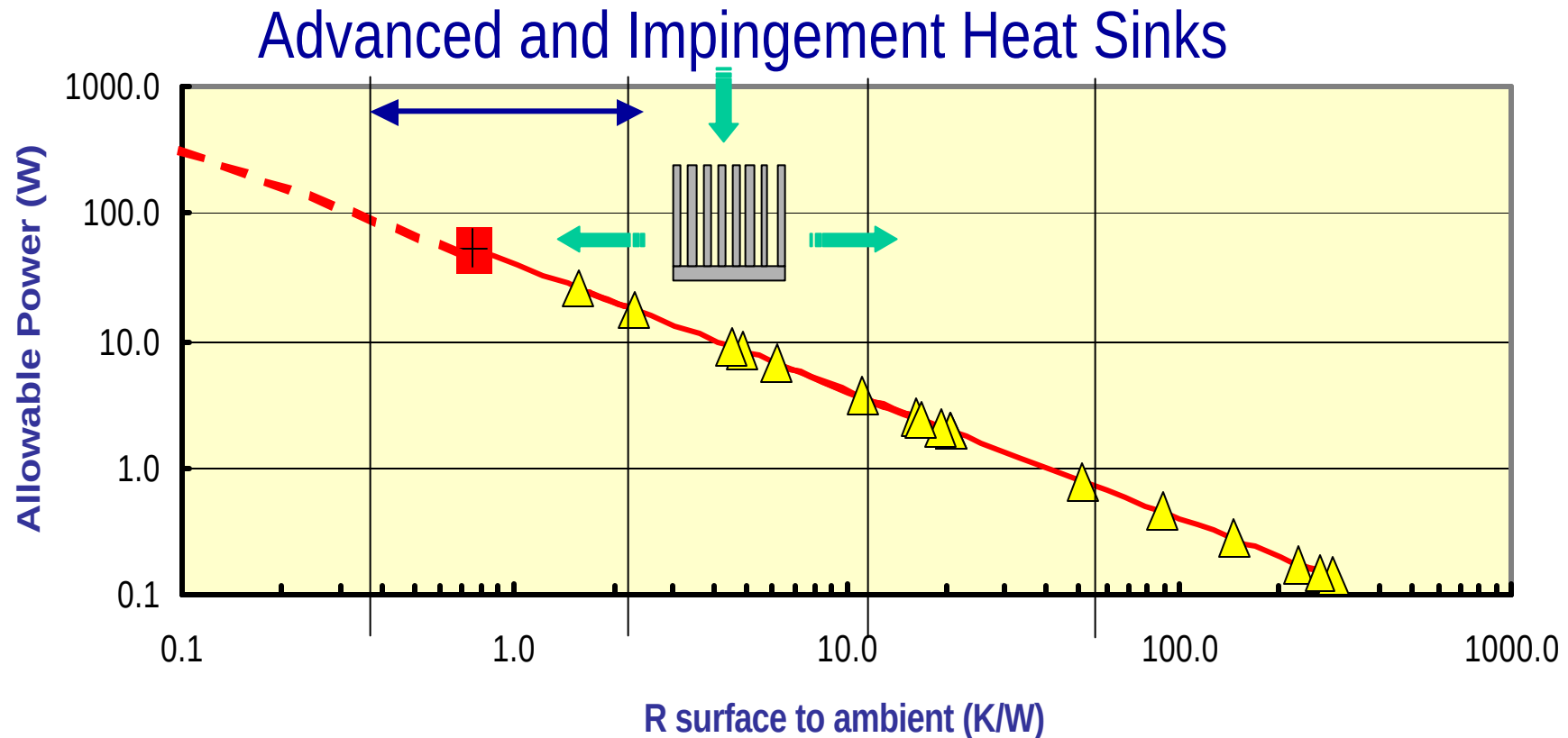


$S/t=2$ Bypass=1



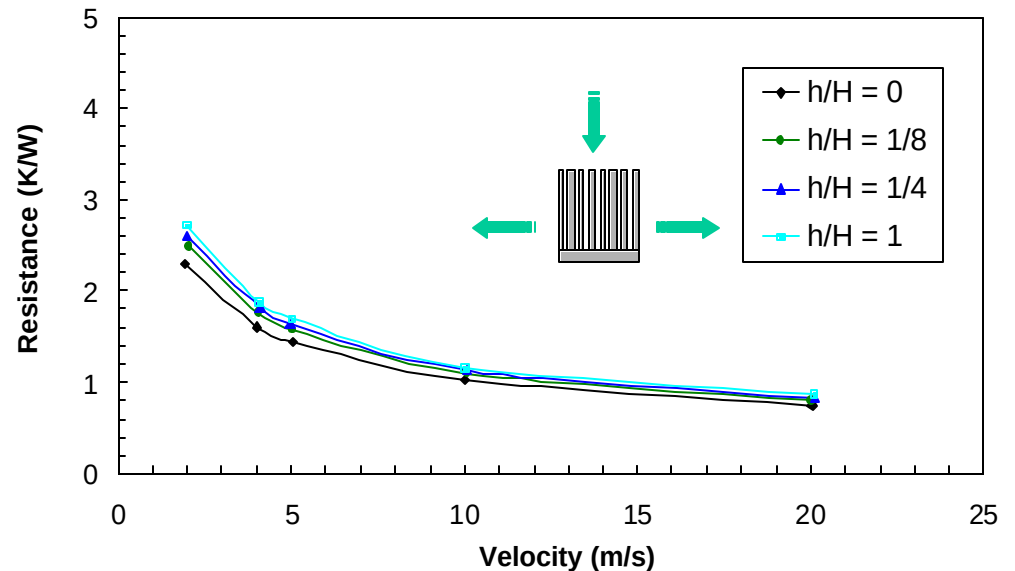
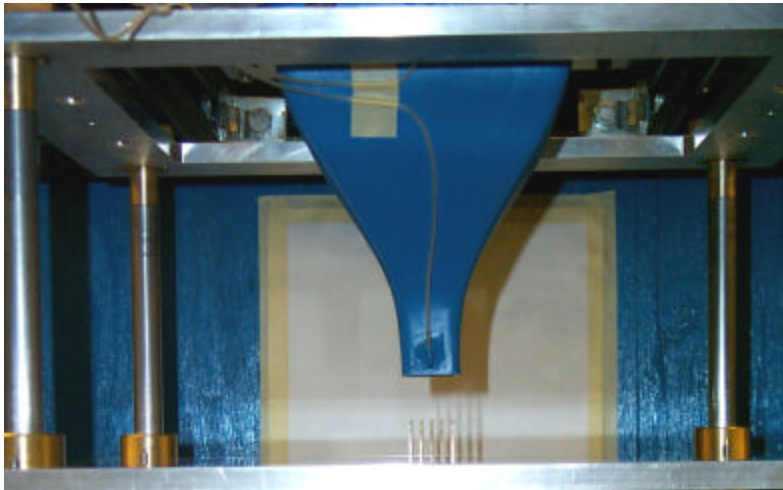
Forced Convection: Pin Fin Heat Sinks in Impinging Flow

Nominal $T_s - T_a = 40^\circ\text{C}$ Planform area = $2.5 \times 2.5 \text{ cm}^2$





Forced Convection: Pin Fin Heat Sinks in Impinging Flow



- ◆ Aggressive Jet Impingement over large area Pin Fin Heat Sinks will achieve R of order 0.2 to 0.5 K/W
- ◆ Jet impingement is a viable strategy for high end processors, but it requires the creation of significant pressure head

- ◆ Fully ducted high aspect ratio plate and pin fin heat sinks in parallel flow offer similar thermal performance.



FUTURE PROSPECTS

- In the past 20 years, we have learned much about the physics of air cooling.
- We know that convective flows in electronics systems exist in regimes that are difficult to model accurately, nevertheless
- CFD simulation has had a major impact in the design process, despite many unanswered questions still.
- The prospects of achieving R less than 0.1 K/W with air are not promising, as long as velocity/acoustic requirements continue to constrain.
- The next five years will either see a migration to liquid or liquid-assisted systems, or the adoption of new circuit technology. We CANNOT continue on the same power growth curve, something has to give.



ACKNOWLEDGEMENTS

This work would not have been possible without the support of the my research partners over the past 15 years. With my sincere appreciation to the following:

AMD

SRC

Intel

NSF

Cisco

DOE

Texas Instruments

AFOSR

Raytheon Defense

IBM

Compaq

Honeywell

Aavid